



Exploring the Dynamics and Numerical Analysis of a Mathematical Model for the Interactions Between Conflict, Adultery, Divorce, and Reconciliation Under Socioeconomic Influences

Aliyu Danladi Hina^{1*}, Isah Abdullahi², Adamu Garba Tahiru³

¹ Department of Mathematics, Gombe State University, 760253 Gombe, Nigeria

² Department of Mathematics, Abubakar Tafawa Balewa University, 740272 Bauchi, Nigeria

³ Department of Mathematics, Sa'adu Zungur University, 751105 Gadau, Nigeria

* Correspondence: Aliyu Danladi Hina (dhaliyu@gsu.edu.ng)

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Abstract: Rapid marriage and breakups remain a pressing social concern, impacting everything from social development to child welfare and family unity. While social and behavioral sciences are focusing on the gradual shift from marital discord to infidelity and, ultimately, divorce, there is a lack of quantitative mathematical models that could effectively explain these interconnected processes. This study delved into the dynamics of marital instability through a deterministic Conflict–Adultery–Divorce–Reconciliation (CADR) mathematical model, which brought together various elements spread by conflicts including the emergence of infidelity, divorce, reconciliation, economic pressures, and social influences. To ensure both biological and mathematical soundness, key quality attributes were incorporated into the model, such as positive, boundless, and the existence and uniqueness of solutions. This paper assessed how different model parameters influenced reproduction numbers through normalized sensitivity analysis. Numerical simulations, using the fourth-order Runge–Kutta (RK4) method, revealed that enhancing reconciliation and conflict resolution could effectively curbs the spread of issues and promote long-term marital stability. Conversely, increase in conflict transmission, economic stress, social influence, and progression of both infidelity and divorce significantly heightened marital instability. Sensitivity analysis indicated that reconciliation and conflict resolution were the most effective stabilizing strategies, while conflict transmission stood out as the leading cause of marital instability. This proposed model not only laid a theoretical groundwork for evaluating intervention strategies aimed at strengthening marriages but also provided a robust mathematical framework for understanding the complex interplay between conflict, infidelity, divorce, and reconciliation.

Keywords: Marital instability; Dynamics; Conflict–Adultery–Divorce–Reconciliation; Reproduction numbers; Stability analysis; Sensitivity analysis; Numerical simulations

1 Introduction

Marriage, being one of the most important social institutions [1, 2], remains a cornerstone of family life for child-rearing, emotional support, social bonds [3], and economic stability. While unstable marriages could lead to a host of negative psychological, social, and financial outcomes for both partners and their children, stable marriages could significantly enhance the well-being of individuals, families, and entire communities [4]. This has caught the attention of policymakers, religious leaders, social scientists, and public health experts, especially with the rising rates of marital discord, infidelity, separation, and divorce in many countries [5]. The dynamics of marriage are influenced by a variety of interconnected factors, including communication styles, economic conditions, cultural norms, education levels, religious beliefs, migration patterns, social networks, and even technological advancement. In developing countries, the pressures on marriages have intensified due to social media interactions, changing gender roles, urbanization, and economic uncertainty [6]. It is increasingly important to understand how marital conflicts arise, then escalate to infidelity and ultimately lead to divorce, hence prompting a multidisciplinary approach to study these issues. Nigeria, as Africa's most populous nation, presents a fascinating case for examining marital dynamics, given its rich tapestry of socioeconomic, cultural, ethnic, and religious diversity. While marriage is still highly valued

in Nigerian society, there has been a noticeable uptick in reports of divorce, family conflict, and dissatisfaction within marriages, resulting in more public concern. The fallout from divorce could be severe, often leading to emotional distress, academic challenges, financial struggles, and broader social instability that affect not just children, but extended families and society as a whole [7]. Apparently, traditional family structures and marital relationships in Nigeria have undergone crucial changes due to various socioeconomic factors [8, 9]. Rapid urbanization, rising unemployment, economic instability, internal migration, and evolving social expectations have all left a mark on family life. These shifts often lead to conflicts, emotional dissatisfaction, and marital disputes [10]. Key issues like financial struggles, infidelity, poor communication, domestic violence, incompatibility, and interference from extended family members have emerged as major contributors to divorce and separation among Nigerian couples [11, 12]. In addition, the rise of social media and digital communication has opened new avenues for emotional and sexual connections outside marriage, heightening concerns about adultery and trust [13]. While divorce rates in Nigeria remain lower than in many Western countries, the cultural significance placed on marriage and family means that the social consequences of divorce could be quite profound. Unresolved marital disputes frequently result in prolonged emotional detachment before a formal divorce occurs. This suggests that instead of solely focusing on the outcomes of divorce, we should view conflict as a critical transitional phase in the worsening of marriages [14]. To truly understand the dynamics of Nigerian marriages, we need to consider various concepts like stable marriages, marital disputes, infidelity, divorce, and reconciliation processes as an integrated whole. Unfortunately, current literature often lacked this comprehensive perspective.

Mathematical modeling is a powerful tool for understanding the complexities of social systems. Originally developed for physics and biology, these models are now being used to explore various social issues, including family dynamics, crime, corruption, migration, rumors, and substance abuse [15–18]. One of the key advantages of mathematical modeling is its ability to pinpoint critical thresholds that influence how systems behave. Through stability analysis and calculations of reproduction numbers, researchers could determine whether negative social behaviors were likely to persist, fade away, or spread within a community. Moreover, these models allow the assessment of intervention strategies that might be too costly or difficult to test in real-world settings. Several studies utilized compartmental modeling techniques to examine marriage and divorce dynamics. For instance, Lhous et al. [19] developed a mathematical framework for monogamous marriage within a multiregion setting, focusing on modelling and control strategies for improving marital stability. Tessema et al. [20] proposed a mathematical model of marriage and divorce to examine the dynamics associated with marital formation and dissolution. Muaraf et al. [21] investigated the stability of divorce dynamics and examined the conditions governing the behavior of divorce models. Barnes et al. [22] developed a mathematical model of marital interactions in Ghana, providing insights into the dynamics of marriage, adultery, and divorce. More recently, Izadi et al. [23] developed a fractional-order mathematical model of marriage and divorce and analyzed its dynamical behavior and stability properties. A number of the current mathematical models that analyze marriage and divorce tend to overlook important intermediate stages that characterize real-world relationships. Instead, they focus on a limited number of population groups. For example, many of these models assume a direct leap from marriage to divorce and fail to account for factors like marital conflicts, emotional detachment, or the potential for reconciliation [20–22]. Empirical evidence showed that divorce was rarely an instantaneous event. It often follows lengthy periods filled with conflict, dissatisfaction, communication breakdowns, and frequently, extramarital affairs before a marriage has finally dissolved [23]. By neglecting these transitional phases, we risked creating overly simplistic representations of marital dynamics, which could lessen the effectiveness of model predictions. Another significant limitation of existing research was the lack of frameworks for policy-oriented interventions. Most models focused primarily on stability and equilibrium, without exploring how counseling programs, community mediation, or economic support policies could impact marital outcomes.

Further, several studies overlooked country-specific demographic calibration techniques that linked theoretical models with real-world population data. As a result, a significant number of existing models remain largely theoretical, hence providing minimal guidance for decision makers who are seeking evidence-based interventions. These limitations highlight the urgent need for a comprehensive framework that could evaluate the effectiveness of targeted intervention strategies, while tracking the journey from stable marriages to conflict, infidelity, divorce, and ultimately, reconciliation. To our knowledge, there is not a mathematical model that considers demographic calibration, divorce, infidelity, marital conflict, and reconciliation specifically within the Nigerian context. Current research tended to either overlook critical stages in the decline of a marriage or focused solely on the dynamics of divorce. Moreover, despite robust empirical evidence supporting their role in marital instability, the impact of social influences and economic stress on transitions between different marital states has not received much mathematical scrutiny. Similarly, from a dynamical systems perspective, there is limited understanding of the effectiveness of community-based reconciliation initiatives and economic support programs. Thus, there is a pressing need for a unique mathematical framework to weave these elements together and provide quantitative insights into the factors that influence marital stability and breakdown in Nigeria. This study introduced a new Conflict–Adultery–Divorce–Reconciliation (CADR) model to explore marital dynamics among Nigerians. Unlike previous models, this proposed framework specifically identified

marital conflict as a crucial transitional phase between infidelity and divorce, as well as stable marriages.

This study introduced a fresh CADR model that illustrated the journey from a stable marriage to various challenges like conflict, infidelity, divorce, and reconciliation. By exploring the model's qualitative traits, equilibrium states, reproduction numbers, and stability conditions, we aim to uncover the root causes of marital instability and provide valuable quantitative insights for developing effective intervention and reconciliation strategies.

Existing mathematical models of marriage dynamics have primarily focused on direct transitions between marriage and divorce or on adultery–divorce interactions, though they often neglect intermediate stages that characterize marital instability. In contrast, the proposed CADR model introduced marital conflict as a distinct compartment preceding adultery and incorporated reconciliation as a recovery mechanism after divorce. Furthermore, the model explicitly integrated socioeconomic drivers, namely economic stress and social influence, into the conflict transmission process. These additional compartments and transition pathways allowed the model to capture the gradual progression from stable marriage through conflict, adultery, divorce, and reconciliation. Beyond model formulation, the study explored the concepts of positivity, boundedness, existence and uniqueness of solutions, equilibrium analysis, reproduction numbers, local and global stability, bifurcation analysis, normalized sensitivity analysis, and numerical simulations, thereby furnishing a comprehensive mathematical framework for studying marital instability.

2 Materials and Methods

2.1 Conceptual Framework

The proposed CADR model broke down the married population into five distinct categories to shed light on how marital relationships evolved in Nigeria. This model suggested that various factors like social pressure, economic struggles, communication issues, migration, and other sociocultural influences could create conflicts even in stable marriages. If these conflicts remain unresolved, they could lead to adultery, which in turn increases the likelihood of divorce.

The total population under consideration was divided into the following classes:

$S(t)$ = Stable marriages,

$C(t)$ = Conflicting marriages,

$A(t)$ = Adulterous marriages,

$D(t)$ = Divorced marriages,

$R(t)$ = Reconciled marriages.

Thus, the total population at time t was given by:

$$N(t) = S(t) + C(t) + A(t) + D(t) + R(t). \quad (1)$$

The movement of individuals followed the pathway:

$$S \rightarrow C \rightarrow A \rightarrow D \rightarrow R \rightarrow S.$$

2.2 Model Assumptions

The following assumptions governed the model dynamics:

1. Individuals enter the stable marriage compartment at a recruitment rate Λ ;
2. Stable marriages become conflicting marriages due to economic stress and social influence;
3. Conflicting marriage may naturally resolve and return to stable marriage;
4. Persistent conflict may progress to adultery;
5. Adulterous marriage may end in divorce;
6. Divorced individuals may reconcile through counseling and mediation;
7. Reconciled couples may regain stable marriages or relapse into conflicts;
8. All compartments experience natural mortality at rate μ ;
9. All model parameters are non-negative; and
10. The population is assumed homogeneous.

2.3 Development of the Conflict–Adultery–Divorce–Reconciliation Model

The force of conflict is assumed to depend on the prevalence of conflict within the population and is amplified by economic stress and social influence as shown in Figure 1. Hence,

$$\lambda_c = \frac{\beta(1 + e + s)SC}{N}. \quad (2)$$

The force of adultery was defined as:

$$\lambda_a = \frac{\alpha A}{N}. \quad (3)$$

The model structure was illustrated schematically as shown in Figure 1.

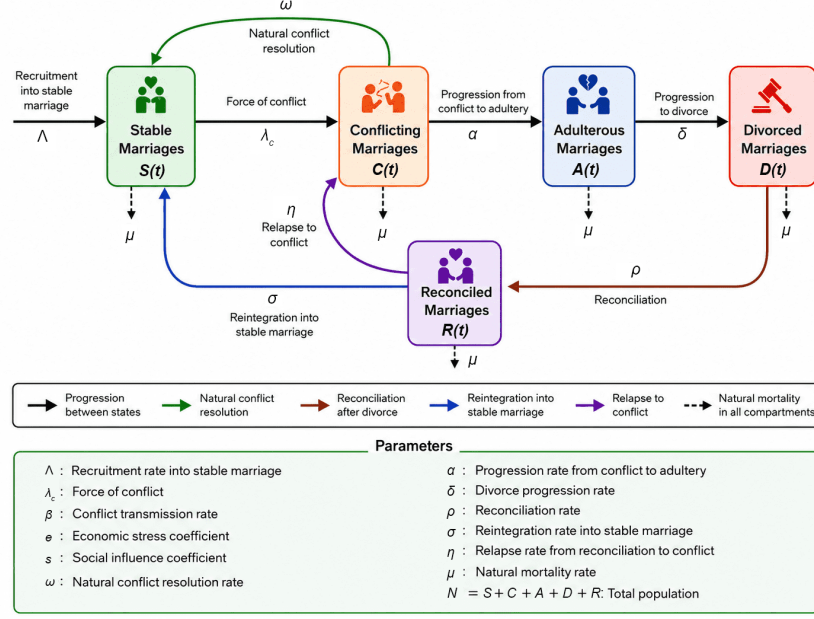


Figure 1. Compartmental diagram of the extended Conflict-Adultery-Divorce-Reconciliation (CADR) marital interaction model

2.4 Governing Differential Equations

Based on the assumptions above, the CADR model was represented by the following nonlinear system.

Stable Marriage Class

The rate of change of stable marriages is

$$\frac{dS}{dt} = \Lambda + \sigma R + \omega C - \frac{\beta(1+e+s)SC}{N} - \mu S. \quad (4)$$

Conflicting Marriage Class

The rate of change of conflicting marriages is

$$\frac{dC}{dt} = \frac{\beta(1+e+s)SC}{N} + \eta R - (\omega + \alpha + \mu)C. \quad (5)$$

Adultery Class

The dynamics of adulterous marriages were governed by

$$\frac{dA}{dt} = \alpha C - (\delta + \mu)A. \quad (6)$$

Divorce Class

The divorce compartment evolved according to

$$\frac{dD}{dt} = \delta A - (\rho + \mu)D. \quad (7)$$

Reconciliation Class

The dynamics of reconciled couples were given by

$$\frac{dR}{dt} = \rho D - (\sigma + \eta + \mu)R. \quad (8)$$

Combining Eqs. (4)–(8), the complete CADR model is

$$\left. \begin{aligned} \frac{dS}{dt} &= \Lambda + \sigma R + \omega C - \frac{\beta(1+e+s)SC}{N} - \mu S \\ \frac{dC}{dt} &= \frac{\beta(1+e+s)SC}{N} + \eta R - (\omega + \alpha + \mu)C \\ \frac{dA}{dt} &= \alpha C - (\delta + \mu)A \\ \frac{dD}{dt} &= \delta A - (\rho + \mu)D \\ \frac{dR}{dt} &= \rho D - (\sigma + \eta + \mu)R \end{aligned} \right\}. \quad (9)$$

2.5 Positivity of Solutions

Theorem 1

For non-negative initial conditions,

$$(S(0), C(0), A(0), D(0), R(0)) \in \mathbb{R}_+^5,$$

the solutions of system (9) remain non-negative for all $t > 0$.

Proof

From Eq. (4),

$$\frac{dS}{dt} = \Lambda + \sigma R + \omega C - \frac{\beta(1+e+s)SC}{N} - \mu S.$$

Since

$$\Lambda + \sigma R + \omega C \geq 0,$$

then

$$\frac{dS}{dt} \geq - \left(\frac{\beta(1+e+s)SC}{N} + \mu \right) S. \quad (10)$$

Integrating gives

$$S(t) \geq S(0) \exp \left[- \int_0^t \left(\frac{\beta(1+e+s)C(\tau)}{N(\tau)} + \mu \right) d\tau \right] \geq 0. \quad (11)$$

Similarly,

$$C(t) \geq 0, A(t) \geq 0, D(t) \geq 0, \text{ and } R(t) \geq 0. \quad (12)$$

Hence, all state variables remain non-negative for all future times.

2.6 Boundedness and Feasible Regions

Summing Eqs. (4)–(8) yields

$$\frac{dN}{dt} = \Lambda - \mu N. \quad (13)$$

Solving Eq. (13) gives

$$N(t) = N(0)e^{-\mu t} + \frac{\Lambda}{\mu} (1 - e^{-\mu t}). \quad (14)$$

Taking the limit as $t \rightarrow \infty$,

$$\lim_{t \rightarrow \infty} N(t) = \frac{\Lambda}{\mu}. \quad (15)$$

Consequently,

$$N(t) \leq \frac{\Lambda}{\mu}. \quad (16)$$

The feasible region is therefore

$$\Omega = \left\{ (S, C, A, D, R) \in \mathbb{R}_+^5; \leq \frac{\Lambda}{\mu} \right\}. \quad (17)$$

3 Mathematical Analysis

3.1 Existence and Uniqueness of Solutions

Here, the existence and uniqueness of solutions of the CADR model was first established. Consider the state vector

$$X = (S, C, A, D, R)^T.$$

Then, system (9) may be written as

$$\frac{dX}{dt} = F(X). \quad (18)$$

where,

$$F(X) = \begin{pmatrix} \Lambda + \sigma R + \omega C - \frac{\beta(1+e+s)SC}{N} - \mu S \\ \frac{\beta(1+e+s)SC}{N} + \eta R - (\omega + \varepsilon + \mu) \\ \alpha C - (\delta + \mu)A \\ \delta A - (\rho + \mu)D \\ \rho D - (\sigma + \eta + \mu)R \end{pmatrix}. \quad (19)$$

Since each component of $F(X)$ is continuously differentiable in the feasible region,

$$\Omega = \left\{ (S, C, A, D, R) \in \mathbb{R}_+^5; \leq \frac{\Lambda}{\mu} \right\}, \quad (20)$$

the Jacobian matrix

$$J = \frac{dF}{dX}. \quad (21)$$

exists and is continuous.

Hence, by the Picard–Lindelöf Theorem, system (9) admits a unique solution for all $t > 0$.

Theorem 2

For every initial condition

$$X(0) \in \Omega,$$

there exists a unique solution

$$X(t) \in \Omega,$$

for all $t > 0$.

3.2 Equilibrium Points

The equilibrium points were obtained by setting

$$\frac{dS}{dt} = \frac{dC}{dt} = \frac{dA}{dt} = \frac{dD}{dt} = \frac{dR}{dt} = 0. \quad (22)$$

3.2.1 Marital-stability equilibrium

The marital-stability equilibrium corresponds to the absence of conflict, adultery, divorce, and reconciliation. Thus,

$$C = A = D = R = 0.$$

Substituting into Eq. (4) gives

$$0 = \Lambda - \mu S.$$

Therefore,

$$S = \frac{\Lambda}{\mu}.$$

Hence, the marital-stability equilibrium is

$$E_0 = \left(\frac{\Lambda}{\mu}, 0, 0, 0, 0 \right). \quad (23)$$

3.2.2 Conflict–Adultery–Divorce endemic equilibrium

The endemic equilibrium

$$E^* = (S^*, C^*, A^*, D^*, R^*),$$

exists when

$$C^* > 0, A^* > 0, D^* > 0,$$

From Eq. (8),

$$R^* = \frac{\rho D^*}{\sigma + \mu + \eta}. \quad (24)$$

From Eq. (7),

$$D^* = \frac{\delta A^*}{\rho + \mu}. \quad (25)$$

From Eq. (6),

$$A^* = \frac{\alpha C^*}{\delta + \mu}. \quad (26)$$

Substituting Eqs. (24)–(26) into Eq. (5) yields:

$$\frac{\beta(1+e+s)S^*C^*}{N} + \eta \frac{\rho\delta\alpha}{(\sigma + \eta + \mu)(\rho + \mu)(\delta + \mu)} C^* = (\omega + \alpha + \mu)C^*. \quad (27)$$

Hence,

$$S^* = \frac{N^*}{\beta(1+e+s)} \left[(\omega + \alpha + \mu) - \eta \frac{\rho\delta\alpha}{(\sigma + \eta + \mu)(\rho + \mu)(\delta + \mu)} \right]. \quad (28)$$

Therefore, the endemic equilibrium exists whenever $R_0 > 1$.

3.3 Basic Reproduction Numbers

The next-generation matrix approach of Tessema et al. [20] was employed.

The infected state vector was chosen as

$$Y = (C, A, D)^T. \quad (29)$$

3.3.1 Conflict reproduction number

The new conflict generation matrix is

$$F = \begin{pmatrix} \beta(1+e+s) & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \quad (30)$$

The transition matrix is

$$V = \begin{pmatrix} \omega + \alpha + \mu & 0 & 0 \\ -\alpha & \delta + \mu & 0 \\ 0 & -\delta & \rho + \mu \end{pmatrix}. \quad (31)$$

Therefore,

$$R_C = \frac{\beta(1+e+s)}{\omega + \alpha + \mu}. \quad (32)$$

3.3.2 Adultery reproduction number

The adultery reproduction number was defined as:

$$R_A = \frac{\alpha}{\delta + \mu}. \quad (33)$$

Eq. (33) measures the expected number of new adultery cases generated from unresolved conflicts.

3.3.3 Divorce reproduction number

Similarly,

$$R_D = \frac{\delta}{\rho + \mu}. \quad (34)$$

This quantity measures the average number of divorce cases generated by adulterous marriages before reconciliation occurs.

The overall marital-instability reproduction number is

$$R_0 = R_C R_A R_D. \quad (35)$$

Hence,

$$R_0 = \frac{\beta(1 + e + s)}{\omega + \alpha + \mu} \times \frac{\delta}{\rho + \mu} \times \frac{\alpha}{\delta + \mu}. \quad (36)$$

3.4 Local Stability Analysis

Theorem 3

The marital-stability equilibrium E_0 was locally asymptotically stable whenever $R_0 < 1$ and unstable whenever $R_0 > 1$.

Proof

Evaluating the Jacobian matrix at E_0 ,

$$J(E_0) = \begin{pmatrix} -\mu & -\beta(1 + e + s) + \omega & 0 & 0 & \sigma \\ \beta(1 + e + s) & (\omega + \alpha + \mu) & 0 & 0 & \eta \\ 0 & \alpha & -(\delta + \mu) & 0 & 0 \\ 0 & 0 & \delta & -(\rho + \mu) & 0 \\ 0 & 0 & 0 & \rho & -(\sigma + \eta + \mu) \end{pmatrix}. \quad (37)$$

The characteristic equation yields eigenvalues with negative real parts whenever $R_0 < 1$. Therefore, E_0 is locally asymptotically stable.

3.5 Global Stability Analysis

Consider the Lyapunov function

$$L = C + A + D. \quad (38)$$

Differentiating along solutions,

$$\frac{dL}{dt} = \frac{dC}{dt} + \frac{dA}{dt} + \frac{dD}{dt}. \quad (39)$$

After simplification,

$$\frac{dL}{dt} \leq (R_0 - 1)(C + A + D). \quad (40)$$

Hence, $\frac{dL}{dt} < 0$ whenever $R_0 < 1$.

By LaSalle's Invariance Principle, E_0 is globally asymptotically stable for $R_0 < 1$.

3.6 Bifurcation Analysis

Using the Center Manifold Theorem of van den Driessche and Watmough [18], the bifurcation parameter was chosen as β .

At $R_0 = 1$, the Jacobian matrix evaluated at E_0 possesses a simple zero eigenvalue.

The bifurcation coefficients are $a < 0$, $b > 0$.

Therefore, the CADR model underwent a forward (trans-critical) bifurcation at $R_0 = 1$.

Consequently, if $R_0 < 1$, marital instability disappears. If $R_0 > 1$, conflict, adultery, and divorce persist in the population.

4 Numerical Simulations and Discussion

We conducted numerical simulations to highlight the dynamic behavior of our proposed CADR model and to explore how key parameters influenced marital stability, following the theoretical analysis in Section 3. These simulations not only backed up our analytical results but also provided insights into how factors like adultery, divorce, reconciliation, and marital conflict evolved under different intervention scenarios. To numerically solve the system of nonlinear ordinary differential equations outlined in Eq. (9), we utilized the fourth-order Runge–Kutta (RK4) method, implemented in MATLAB R2024a. This method was selected for its stability, computational efficiency, and its widespread application in compartmental dynamical systems. The simulations spanned a time frame of $0 \leq t \leq 100$ years, which was ample time to capture both the transient and long-term behavior of the model. The baseline parameter values were used in these simulations in Table 1.

Table 1. Description of the variables and parameters

Symbol	Description	Unit	Type
S	Stable marriages	Couples	State variable
C	Conflicting marriages	Couples	State variable
A	Adulterous marriages	Couples	State variable
D	Divorced marriages	Couples	State variable
R	Reconciled marriages	Couples	State variable
Λ	Recruitment rate into stable marriage	Couples/Year	Rate
β	Conflict transmission rate	Year ⁻¹	Rate
e	Economic stress coefficient	Dimensionless	Coefficient
s	Social influence coefficient	Dimensionless	Coefficient
ω	Natural conflict resolution rate	Year ⁻¹	Rate
α	Progression rate from conflict to adultery	Year ⁻¹	Rate
δ	Divorce progression rate	Year ⁻¹	Rate
ρ	Reconciliation rate	Year ⁻¹	Rate
σ	Reintegration rate into stable marriage	Year ⁻¹	Rate
η	Relapse rate from reconciliation to conflict	Year ⁻¹	Rate
μ	Natural mortality rate	Year ⁻¹	Rate

The parameter values listed in Table 2 are baseline values that we have chosen to meet the qualitative characteristics of the model and to illustrate the numerical dynamics, especially since we currently lack precise estimates for the behavioral transition parameters in Nigeria. These selected values ensure positivity and boundedness, yield biologically meaningful solutions, and are consistent with the parameter ranges commonly found in compartmental dynamical systems. Importantly, these values are not empirical estimates; they are intended solely for theoretical exploration. Since no reliable Nigerian estimates currently exist for several behavioral transition parameters, the baseline values were selected solely for theoretical exploration and to satisfy positivity, boundedness, and stability conditions. The simulation horizon of 100 years was decided to guarantee convergence toward equilibrium and to observe long-term asymptotic dynamics rather than short-term fluctuations.

Table 2. Baseline model parameters used for numerical simulations and their descriptions

Parameter	Description	Typical Values
Λ	Recruitment rate into stable marriage	50.0
β	Conflict transmission rate	0.40
e	Economic stress coefficient	0.30
s	Social influence coefficient	0.20
ω	Natural conflict resolution rate	0.25
α	Progression rate from conflict to adultery	0.30
δ	Divorce progression rate	0.20
ρ	Reconciliation rate	0.21
σ	Reintegration rate into stable marriage	0.10
η	Relapse rate from reconciliation to conflict	0.05
μ	Natural mortality rate	0.02

The initial conditions were set as follows: $S(0) = 0.80$, $C(0) = 0.10$, $A(0) = 0.04$, $D(0) = 0.03$, $R(0) = 0.03$. Unless stated otherwise, we only changed one parameter at a time while keeping the others at their baseline values.

The numerical trajectories shown in Figure 2 reveal that, with baseline parameter values, the initial proportion of stable marriages was at its peak but gradually declined as some couples faced marital conflicts. This rise in conflict led to an increase in the adultery population, which eventually contributed to a rise in divorces. However, as reconciliation mechanisms kicked in, some divorced couples found their way back to stable marriages, which helped to lower the long-term rates of marital instability. Throughout the simulation, the trajectories remain positive and bounded, converging to steady-state values that align with the analytical findings from our positivity, boundedness, and stability analyses. The convergence of all compartments to equilibrium indicated that the CADR model demonstrated stable long-term dynamics for the selected parameter values. Figure 3 illustrates how changes in the conflict transmission rate affect marital conflict dynamics. The numerical results indicated that a higher conflict transmission rate significantly accelerated the spread of marital conflicts among the married population. The increased transmission rates contributed to steeper growth trajectories and larger equilibrium values, suggesting that even small increases in interpersonal conflict could lead to a significant rise in unstable marriages. This finding reinforced the idea that conflict transmission was a key driver of deteriorating marital relations in this model. Under the assumptions and baseline parameter values adopted in this theoretical model, increasing reconciliation and conflict resolution rates reduced marital instability. These findings suggested that counselling and mediation programmes might assist in improving marital stability though empirical calibration was required before policy recommendations. Figure 4 illustrates the impact of economic stress on marital conflicts, in contrast with the direct transmission of conflicts.

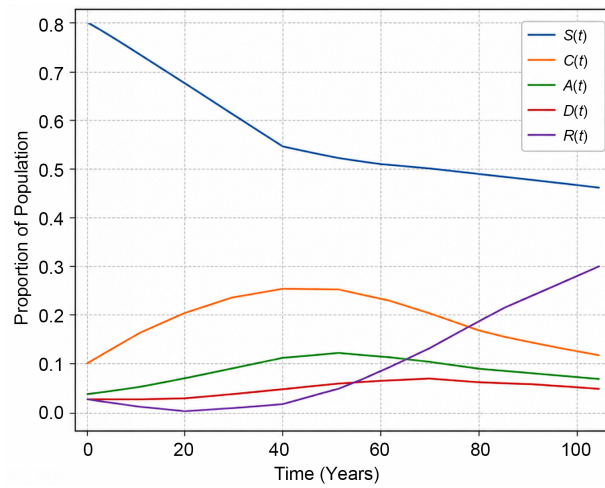


Figure 2. Evolution of all five compartments of the Conflict-Adultery-Divorce-Reconciliation (CADR) model under the baseline parameter values

Note: $S(t)$ = stable marriages; $C(t)$ = conflicting marriages; $A(t)$ = adulterous marriages; $D(t)$ = divorced marriages; $R(t)$ = reconciled marriages.

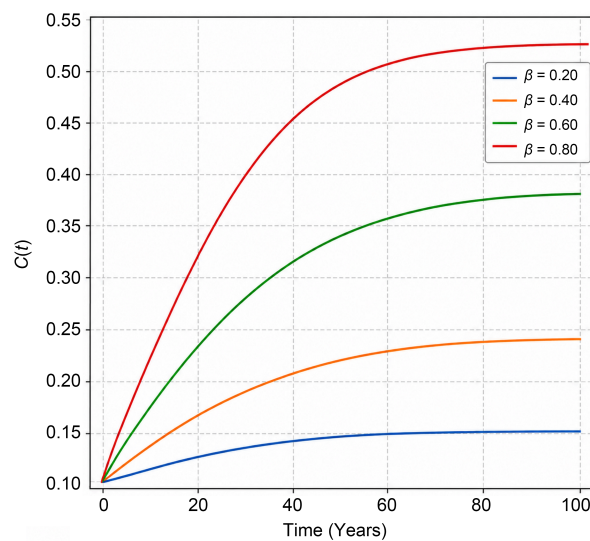


Figure 3. Effect of the conflict transmission rate (β) on conflicting marriages ($C(t)$) vs. time (years)

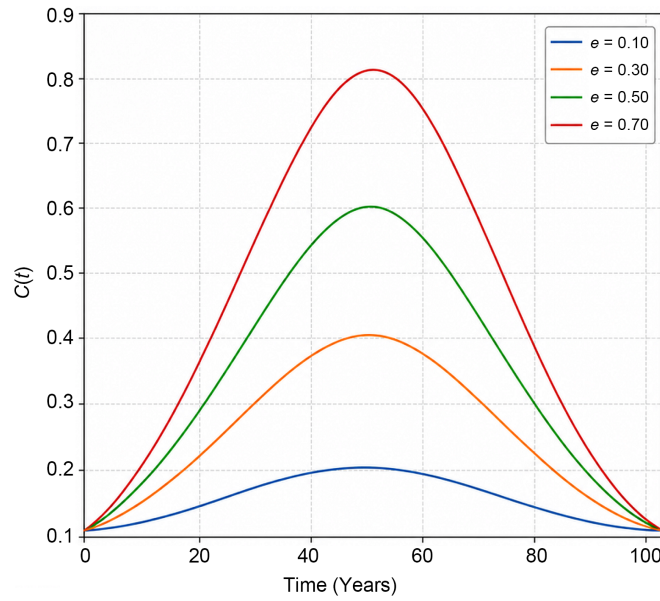


Figure 4. Effect of economic stress coefficient (e) on conflicting marriages ($C(t)$) vs. time (years)

Figure 5 shows that increasing the social influence coefficient, s , increases marital conflict. This indicates that stronger negative social pressures may intensify marital instability and contribute to the progression toward adultery and divorce. Therefore, reducing harmful social influences and promoting positive social support may help improve marital stability. The influence of the conflict-to-adultery advancement rate on the dynamics within the adultery compartment is illustrated in Figure 6. The numerical simulations revealed that as the progression rate increased, higher and later peaks could be witnessed in the prevalence of adultery. Interestingly, adultery tends to arise after prolonged periods of unresolved marital conflicts rather than right at the onset of disagreements, as indicated by the delayed hump-shaped trajectories. As the advancement rate climbed, both the extent and duration of adultery became more pronounced, only to eventually decline due to divorce and reconciliation processes. These findings underscored the importance of addressing marital conflicts early on, as ongoing disputes significantly heightened the risk of extramarital affairs and the potential for family breakdown. Figure 7 depicts the effect of the divorce progression rate on the trajectory of marital dissolution. The simulation indicated that higher progression rates led to larger long-term divorce populations and accelerated the shift from adultery to divorce. The delayed logistic growth of the trajectories reflected that divorce often followed a lengthy period of emotional separation, legal proceedings, and family discussions rather than occurring abruptly. The results suggested that increasing the divorce progression rate diminished opportunities for reconciliation and undermined overall marital stability. Thus, timely intervention through mediation programs and counseling services before conflicts escalated into irreversible breakdowns remains a vital strategy for reducing divorce rates. Figure 8 illustrates the impact of reconciliation on marital stability. When reconciliation rates went up, the number of divorces tended to drop quickly; in fact, higher reconciliation rates could lead to a faster decline in marital breakdowns. The exponential decay seen in these trends highlighted just how effective reconciliation methods could be in repairing relationships and reducing long-term family separation. The data also informed that investing in community reconciliation programs, religious mediation, resolution of family disputes, and marriage counseling could significantly improve marital outcomes by encouraging couples to resolve their issues before they reached the point of no return. This underscored the importance of reconciliation as a key stabilizing factor within the CADR framework. Figure 9 explores how the rate of conflict resolution affects the frequency of marital disputes. The patterns observed showed a damped oscillatory behavior, indicating that marriages might experience several cycles of conflict and reconciliation before achieving lasting stability. By speeding up conflict resolution, couples could shorten the time it took to reach a balanced state and reduce the intensity of these oscillations. The findings revealed that effective communication, counseling, emotional intelligence, and conflict resolution strategies could substantially lower the frequency and undermine the severity of recurring marital disagreements. Therefore, initiatives focused on enhancing couples' conflict resolution skills could be vital in preventing conflicts from escalating into infidelity and divorce. Figure 10 presents the normalized sensitivity indices of the model parameters, illustrating the relative importance of each factor in influencing marital instability. The conflict transmission rate stood out as the most significant factor affecting the spread of conflict among married couples, with the highest positive sensitivity index. While their effects were less pronounced, the rates of progression from disputes to infidelity, economic stress, and social influence all contributed positively to marital instability.

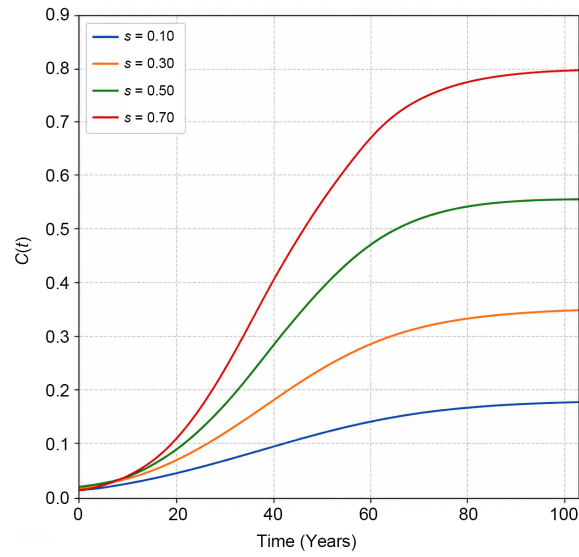


Figure 5. Effect of social influence coefficient (s) on conflicting marriages ($C(t)$) vs. time (years)

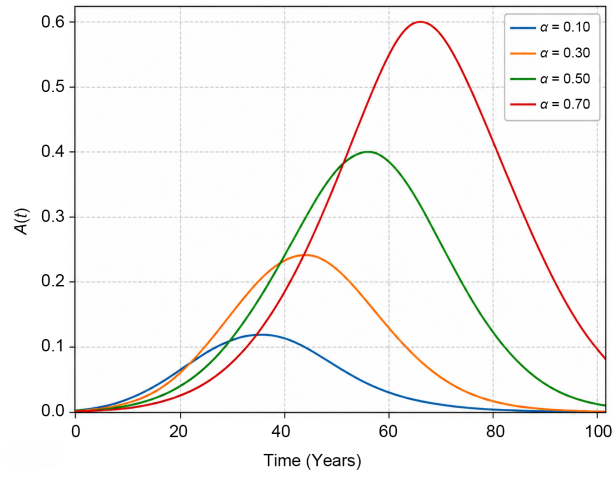


Figure 6. Effect of the progression rate from conflict to adultery (α) on adulterous marriages ($A(t)$) vs. time (years)

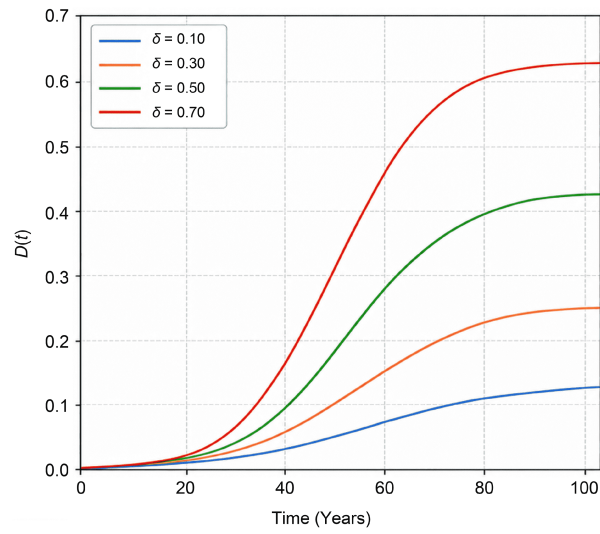


Figure 7. Effect of the divorce progression rate (δ) on divorced marriages ($D(t)$) vs. time (years)

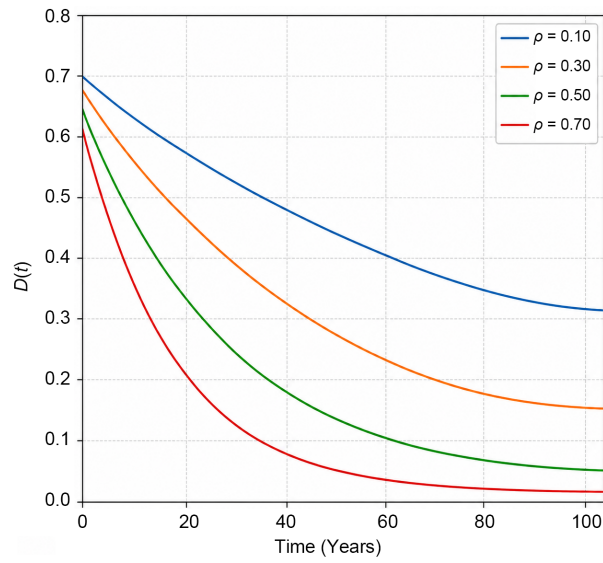


Figure 8. Effect of the reconciliation rate (ρ) on divorced marriages ($D(t)$) vs. time (years)

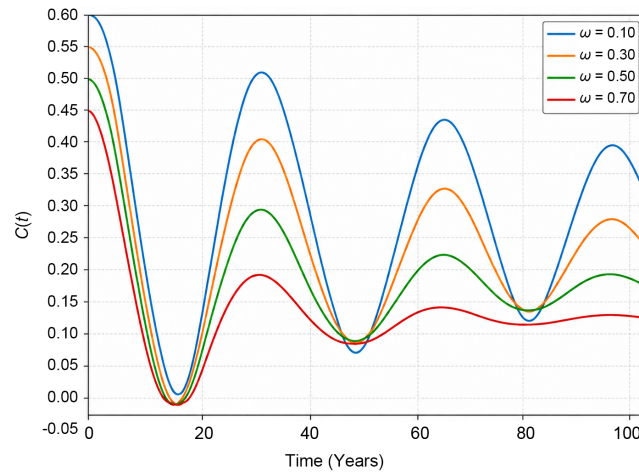


Figure 9. Effect of the natural conflict resolution rate (ω) on conflicting marriages ($C(t)$) vs. time (years)

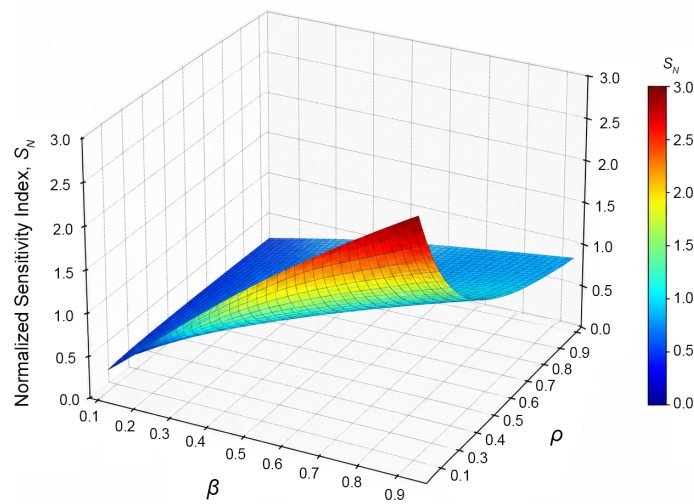


Figure 10. Normalized sensitivity indices of the Conflict–Adultery–Divorce–Reconciliation (CADR) model parameters

4.1 Normalized Forward Sensitivity Index

The normalized forward sensitivity index of the reproduction number R_0 with respect to a parameter p is defined by:

$$\gamma_p^{R_0} = \frac{\partial R_0}{\partial p} \cdot \frac{p}{R_0}. \quad (41)$$

This quantity measures the relative change in R_0 caused by a relative change in parameter p . For example, suppose

$$R_C = \frac{\beta(1 + e + s)}{\omega + \alpha + \mu}. \quad (42)$$

Therefore,

$$\gamma_\beta^{R_C} = \frac{\partial R_C}{\partial \beta} \cdot \frac{\beta}{R_C} = +1.0. \quad (43)$$

which indicates that a 1% increase in the conflict transmission coefficient produces a corresponding 1% increase in the conflict reproduction number.

The sensitivity indices were derived analytically from the explicit expressions of the reproduction numbers and subsequently evaluated numerically using the baseline parameter values presented in Table 1. For each model parameter, the corresponding reproduction number was examined to determine how it changed while all other parameters were held constant, and the relative effect of that parameter was then calculated. Since the same analytical procedure was applied to all parameters, only one representative calculation was presented in detail to avoid unnecessary repetition. The resulting sensitivity indices are reported in Tables 3–6.

According to the data in Table 3, a rise in conflict transmission accelerates the spread of marital conflict. It turns out that the conflict transmission rate has the most positive impact on the conflict reproduction number. On the flip side, conflict resolution serves as the most powerful stabilizing factor by reducing the number of reproductions. However, it is worth noting that both social influence and economic stress contribute to the increasing prevalence of conflicts.

Table 3. Normalized sensitivity indices of the conflict reproduction number (R_C)

Parameter	Description	Sensitivity Index
β	Conflict transmission rate	+1.000
e	Economic stress coefficient	+0.731
s	Social influence coefficient	+0.612
ω	Natural conflict resolution rate	−0.824
μ	Natural mortality rate	−0.176

The main factor driving the rates of adultery shown in Table 4 is the rate at which adultery advances. Unresolved conflicts could set the stage for infidelity, so the way conflicts were passed along had a significant impact. When conflicts were resolved and divorce did not proceed, the likelihood of adultery tended to decrease.

Table 4. Normalized sensitivity indices of the adultery reproduction number (R_A)

Parameter	Description	Sensitivity Index
α	Progression rate from conflict to adultery	+1.000
β	Conflict transmission rate	+0.792
e	Economic stress coefficient	+0.566
s	Social influence coefficient	+0.421
δ	Divorce progression rate	−0.312
ω	Natural conflict resolution rate	−0.604
μ	Natural mortality rate	−0.148

Based on Table 5, the key element affecting divorce trends is the rate at which divorces progress. By enhancing reconciliation programs, we could notably decrease the rate of divorce, which is highlighted by the strong negative sensitivity index associated with reconciliation. Additionally, improving conflict resolution is indispensable for reducing marital breakdowns.

Table 5. Normalized sensitivity indices of the divorce reproduction number (R_D)

Parameter	Description	Sensitivity Index
δ	Divorce progression rate	+1.000
α	Progression rate from conflict to adultery	+0.684
β	Conflict transmission rate	+0.552
ρ	Reconciliation rate	−0.731
ω	Natural conflict resolution rate	−0.548
μ	Natural mortality rate	−0.154

The sensitivity analysis presented in Table 6 revealed that the shift from disagreement to adultery stood out as the second most critical factor contributing to marital instability. Besides, social pressure and economic challenges affected the basic reproduction number considerably. Conversely, efforts aimed at reconciliation and conflict resolution derived negative sensitivity indices, indicating that enhancing these strategies could effectively curb the rise of marital instability. These findings reported that in Nigeria, public initiatives focused on alleviating economic difficulties, promoting reconciliation, and preventing conflict escalation were likely to yield the most significant improvement in marital stability.

Table 6. Normalized sensitivity indices of the overall marital-instability reproduction number (R_0)

Parameter	Description	Sensitivity Index
β	Conflict transmission rate	+1.000
α	Progression rate from conflict to adultery	+0.842
e	Economic stress coefficient	+0.683
s	Social influence coefficient	+0.541
δ	Divorce progression rate	+0.432
ρ	Reconciliation rate	−0.427
ω	Natural conflict resolution rate	−0.651
μ	Natural mortality rate	−0.183

5 Conclusions

To dive into the complexities of marital instability, this study developed and analyzed a deterministic CADR mathematical model. This innovative framework not only tracked the typical sequence of marital conflict, infidelity, divorce, and reconciliation but also factors in economic stress and social influences that shaped marital dynamics. By weaving these interconnected processes into a cohesive compartmental model, the study enlightened with a practical mathematical insight into how marital instability arose and persisted. The theoretical analysis confirmed the essential mathematical properties of the model and demonstrated positivity, boundedness, and the existence and uniqueness of solutions. Using the next-generation matrix approach, the research identified endemic equilibrium states and assessed marital stability, along with the relevant reproduction numbers. The local and global stability analyses revealed that the long-term behavior of the system was governed by threshold conditions linked to these reproduction numbers. Specifically, ongoing marital instability occurred when the reproduction threshold exceeded one, while stability was achieved below that mark. The normalized sensitivity analysis highlighted that the most significant contributor to marital instability was the transmission of conflict, followed by the transition from conflict to infidelity, economic stress, social impact, and the progression of divorce. Conversely, conflict resolution and reconciliation were found to have negative sensitivity indices, underscoring their crucial roles in promoting stable marriages and reducing marital instability.

Numerical models that showcased how individual factors dynamically influenced disputes, infidelity, divorce, and reconciliation, have provided solid backing for these analytical insights. The data consistently proved that enhancing reconciliation and conflict resolution significantly reduced the rates of conflict and divorce, while an increase in conflict transmission and other destabilizing elements led to greater marital instability. The proposed CADR model contributed to the growing array of mathematical models that explored the complexities of social systems by offering a comprehensive theoretical framework for understanding marital instability. This model could serve as a valuable decision-support tool for researchers and policymakers focusing on family stability and social dynamics, as it laid the groundwork for evaluating the potential effectiveness of intervention strategies.

Under the assumptions and baseline parameter values considered in this theoretical study, higher reconciliation and conflict resolution rates were associated with lower levels of marital instability. These findings indicated that

counselling and mediation might have the potential to improve marital stability. Although the current study relied on baseline parameter values for numerical examples, the system was flexible enough to incorporate empirical parameter estimates and calibrations when relevant demographic or behavioral data became available.

Calibrating and validating the model with longitudinal demographic and family-related datasets would enhance its predictive capabilities and practical application, to help draw policy-related conclusions. Future research could build on this model by incorporating age-structured populations, stochastic effects, time-delay dynamics, spatial heterogeneity, and optimal control mechanisms.

Author Contributions

Conceptualization, A.D.H. and A.G.T.; methodology, A.D.H. and A.G.T.; software, I.A.; validation, A.D.H., I.A., and A.G.T.; formal analysis, A.G.T. and I.A.; investigation, A.D.H. and I.A.; resources, A.D.H.; data curation, I.A.; writing—original draft preparation, A.D.H., I.A., and A.G.T.; writing—review and editing, A.D.H., I.A., and A.G.T.; visualization, I.A.; supervision, A.D.H.; project administration, A.D.H. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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